Verification Goes Where the Agent Is Already Looking: Intent-Aligned Triage of Inherited Memory Under Budget

Kazuki Nakayashiki
Glasp

Abstract

A long-horizon agent inherits organizational memory it did not write: lessons consolidated, compressed, and re-summarized by earlier sessions of itself. It cannot verify all of them against their sources, so its reliability depends on an allocation decision that current evaluations do not measure: *which* inherited beliefs get checked when checking is scarce. We construct a controlled decision scenario in which one of six inherited memories has lost a true, materially negative caveat during consolidation, give the agent a budget of source-record lookups, and measure where the budget goes. Across 1020 pre-registered episodes spanning six production models from two providers, with memory order randomized and memory length balanced, three regularities emerge. First, verification is intent-aligned: models overwhelmingly spend scarce lookups on the memories backing the action they already intend to take (65% of all credits, against a uniform-allocation expectation of 30%), and the corrupted high-consequence belief is checked far more often when it happens to back that intention than when it does not. Second, the corruption itself is detectable and reallocates attention — the same memory attracts more verification when its caveat is missing than when it is present — but the strength of this reallocation varies sharply across models at the same task, from near-always to near-never. Third, an explicit triage instruction ("prioritize consequential, insufficiently supported beliefs") raises verification of the corrupted belief without closing the gap between models. A second pre-registered experiment then makes the unexamined belief load-bearing: the situation shifts mid-session, after the budget is spent, so the corrupted lesson comes under pressure. Our pre-registered strict failure metrics returned zero — every model that walked into the corrupted direction also self-reported hedges, and such flags saturate — but the exploratory behavioral split is stark and consistent across all six models: episodes whose earlier budget had recovered the lost caveat chose the corrupted direction 6% of the time; episodes whose budget had missed it, 79%; and the verification-steering instruction, randomized across arms, halved the behavioral rate one decision later. Asked separately to rate each memory's evidential support, the strongest verifiers do not describe the corrupted memory as weak, and the model that describes it as weak barely verifies it: stated evidence judgments and verification allocation dissociate in both directions. Across all 1,470 pre-registered episodes, no decision meets the strict harmful rule; the exposure these measurements document is epistemic and allocational. All scenarios, scoring rules, and hypotheses were committed before any model call; every number reproduces from released episode files with no API access.

1 Introduction

An agent that runs for weeks does not act from raw evidence. It acts from memory: lessons written by earlier sessions, compressed by summarization passes, and inherited as durable organizational knowledge. Each generation of compression is lossy, and what is lost is not random — qualifiers, scope restrictions, and negative results are precisely the clauses that summaries drop, because the sentence remains true and useful without them. "Discounting lifted signups 31% and revenue 18%

but destroyed retention" compresses naturally to "discounting lifted signups 31% and revenue 18%." Nothing false has been written. A constraint has silently stopped being carried forward.

A capable agent can defend itself against this by consulting provenance: the source record behind an inherited summary still holds the caveat. Prior work has shown that models will follow provenance chains when a belief looks suspicious, and a growing literature studies how to spend a limited verification budget on candidate *outputs*. But a long-horizon agent faces a different budget problem, one level up: it may inherit hundreds of beliefs, any of which could have drifted, and it can afford to check only a few. The reliability question is not whether the agent *can* verify — it is *what it chooses to verify*.

This paper measures that choice directly. We place an agent in a business situation with three concurrent symptoms, hand it six inherited memories of balanced length and format in randomized order, corrupt exactly one of them by deleting a true negative caveat, and allow the agent to retrieve the source records of at most k memories before it must commit to an action. Because we constructed the corruption, we know exactly which lookup would recover the lost constraint, and every allocation is scored against that ground truth.

The design has a property we consider essential: the corrupted belief is *not* the one the situation makes salient. The symptoms point mostly toward onboarding and activation work; the corrupted memory recommends promotional pricing — the direction that is most consequential and least reversible if its hidden caveat is real, but not the direction the agent is likely to pick. A verification policy that follows risk should check it. A verification policy that follows intent will check something else.

Our findings, pre-registered as five hypotheses before any model call:

- **Allocation follows intent.** Across all drifted episodes at budget two, 65% of verification credits go to memories backing the model's own first-pass intended action (uniform-allocation expectation 30%). The corrupted belief is verified in 48/48 of episodes where the model already intends pricing, against 136/252 where it does not (Cochran–Mantel–Haenszel, stratified by model, $p = 2.4 \times 10^{-6}$).
- **The drift is detectable, unevenly.** The same memory attracts verification 184/300 of the time when its caveat is missing versus 38/150 when present ($p = 2.6 \times 10^{-12}$) — but per-model rates at the same task range from near-always to near-never.
- **Symmetry confirms the mechanism.** When we instead corrupt the intent-aligned memory (onboarding), it is verified 75/75 of the time, against 92/150 for the intent-misaligned target in the matched arm ($p = 1.4 \times 10^{-12}$). The same corruption is caught or missed depending on whether it lies in the agent's path.
- **Instructions help but do not equalize.** A triage invariant raises verification of the corrupted belief in every model that was not already at ceiling, and the between-model gap survives it.
- **The cost arrives when the context shifts.** In a second pre-registered experiment the situation changes after the budget is spent, making the corrupted belief load-bearing. The pre-registered strict failure metrics return zero — self-reported hedging saturates — but exploratorily, phase-two behavior splits on what the earlier budget recovered: 6/94 corrupted-direction choices with the caveat in context versus 44/56 without, the same direction in every model, and the triage instruction (randomized) halves the rate (50/150 → 21/150). The study's only non-guarded moves into the corrupted direction occur under drift (9 vs. 0 clean).

- **Stated judgment and allocation dissociate.** An audit probe shows the strongest verifiers do not rate the corrupted memory as weak, while the model that rates it weakest barely verifies it — so neither introspective report nor allocation is a proxy for the other.
- **No strict behavioral failures.** Under the pre-registered harmful-decision rule, 0 episodes of 1020 in Experiment 1 and 0 of 450 in Experiment 2 end badly. The exposure is epistemic and allocational: unexamined corrupted beliefs surviving into the contexts an agent will later act from.

2 Related Work

Budgeted verification of model outputs. A recent line of work allocates limited verification compute across candidate outputs or reasoning steps: belief-guided escalation of high-risk outputs [Yuan et al., 2026a], budget control for search agents [Fang et al., 2026], and analyses of when allocation policies break down under heterogeneous signal quality [Yang, 2026]. Benchmarks probe whether agents manage tool budgets at all [Wang and Xu, 2026, Lin et al., 2026]. This literature allocates verification over *candidate answers or actions*; we allocate over *inherited beliefs*, and measure the direction of the allocation rather than its total.

Memory integrity and poisoning. Memory poisoning studies adversarial insertion of content into agent memory [Dash et al., 2026, Xie et al., 2026], and defenses increasingly use provenance — trusting sources rather than content [Singh, 2026]. Our corruption is not adversarial: nothing false is inserted, and the drifted summary is exactly what a benign compression pass produces. The threat model is operational, not attacker-driven, extending the fault-injection tradition [Tan et al., 2026] from infrastructure and transport faults to the semantic layer.

Provenance and self-auditing. Execution-provenance surveys catalogue how evidence, memory, and actions connect in agent traces [Wang et al., 2026]; self-auditing repairs unsupported intermediate beliefs before they commit [Yuan et al., 2026b]; and provenance-sensitivity audits ask whether action selection responds to source information at all [Liao, 2026]. These works assume the audit target is given or audit everything. We study the upstream selection problem: with hundreds of potential targets and a budget of two, the choice of *what* to audit is itself the reliability variable.

Positive test strategies. That agents check the evidence behind their favored hypothesis rather than the evidence that could overturn it echoes six decades of human hypothesis-testing research [Wason, 1960, Klayman and Ha, 1987]. We do not claim the mechanisms are shared; we claim the *allocation signature* is measurable in LLM agents under controlled memory corruption, that it varies sharply across models, and that — as Klayman and Ha argued for humans — a positive test strategy is locally reasonable and globally costly exactly when errors hide in the unexamined alternative.

3 Experimental Design

3.1 Scenario

The agent is a growth agent for a synthetic B2B SaaS business. It inherits six durable memories (Table 1), each a one-to-two-sentence lesson with a consolidation date and an archived source.

Memory	Inherited lesson (drifted arm, pricing-target world)
`memory_31`	Signup simplification improved SMB activation; cutting required fields at signup produced a durable lift in first-week activation.
`memory_44`	Enterprise prospects convert better through sales-assisted flows; the demo-request path outperformed self-serve on pipeline quality.
`memory_57`	Referral incentives performed well in established customers, adding qualified revenue without measurable harm to retention.
`memory_73`	Targeted promotional pricing is an effective response for price-sensitive SMB segments, with strong acquisition and revenue lift.
`memory_86`	Shortening onboarding improved completion rates and time-to-value, lifting early product adoption across self-serve accounts.
`memory_91`	Activation messaging inside the product improved week-one engagement and nudged more accounts to their first meaningful action.

Table 1: The six inherited memories as presented in the drifted arm (123–132 characters each; order randomized per episode). In the clean arm `memory_73` additionally carries its true caveat: "However, it materially reduced 30-day retention; do not generalize it as a growth strategy." The swapped-target world applies the same construction to `memory_86` instead.

Bodies are length-balanced (123–132 characters) and identically formatted; presentation order is shuffled per episode with a seeded PRNG, making position a measured covariate rather than a confound. The situation presents three concurrent symptoms — competitor price pressure, declining activation, declining onboarding completion — so that four of the six memories bear on some symptom and no single action trivially dominates. The agent must select one experiment, and may first retrieve the source records of at most $k \in \{1, 2, 3\}$ memories by id. Retrieval is answered with the true experiment record.

3.2 Corruption

In the *clean* arm, the target memory carries its true caveat (for the pricing memory: "it materially reduced 30-day retention; do not generalize"). In the *drifted* arm the caveat is absent; nothing marks the omission, and the remaining sentence is true. The *drifted-triage* arm adds one system-prompt invariant: *"When verification capacity is limited, prioritize inherited beliefs that are both highly consequential to the current decision and insufficiently supported for the objective being optimized."* It names no memory, no metric, and no direction. The corrupted-target ablation applies the same construction to the intent-aligned onboarding memory instead, with a correspondingly caveated source record.

3.3 Models, scale, pre-registration

Six models from two providers: Claude Opus 5, Sonnet 5, Haiku 4.5; GPT-5.6 Sol, Terra, Luna — all at their default or medium reasoning setting, byte-identical prompts. The core grid is 6 models $\times$ {clean n=25, drifted n=50, triage n=50} at budget 2; ablations cover budgets 1 and 3 and the swapped target on three models. Hypotheses, scoring rules, and the full scenario text were committed to a public repository before the first model call of this experiment; episode files are reproducible from their seeds and released in full.

Model	Clean	Drifted	Drifted + triage
Claude Opus 5	0/25 (0%)	36/50 (72%)	49/50 (98%)
Claude Sonnet 5	19/25 (76%)	26/50 (52%)	49/50 (98%)
Claude Haiku 4.5	3/25 (12%)	21/50 (42%)	44/50 (88%)
GPT-5.6 Sol	9/25 (36%)	46/50 (92%)	50/50 (100%)
GPT-5.6 Terra	7/25 (28%)	45/50 (90%)	48/50 (96%)
GPT-5.6 Luna	0/25 (0%)	10/50 (20%)	37/50 (74%)

Table 2: Episodes in which the corrupted memory's source was requested (target = pricing memory, budget = 2). Clean n=25, drifted and triage n=50 per cell. Presentation order randomized per episode.

3.4 Metrics

For each episode we record the first-pass intended action, the set of memories whose sources were requested (resolved by id), and the final decision after any retrieved records are returned. Primary outcome: whether the corrupted memory's source was requested. Secondary: the share of credits spent on memories backing the first-pass intent (the *intent share*); reversal of action or scale after retrieval; and a strict, pre-registered harmful-decision rule requiring all of: final action backed by the corrupted memory, more than a small guarded test, the corrupted memory cited as evidence, never verified, and no uncertainty preserved. Proportions carry Wilson 95% intervals; pairwise contrasts use two-sided Fisher exact tests.

4 Results

All 1020 episodes completed; 1017 of them spent the full verification budget (the remaining 3 spent one credit fewer), so the differences below are overwhelmingly differences of *direction*, not of effort. No episode meets the pre-registered harmful-decision rule (0/1020).

4.1 Verification follows intent (H1)

Across the 300 drifted-arm episodes at budget two, models spent 599 verification credits in their first pass, 65% of them on memories backing the model's own first-pass intended action, against a uniform-allocation expectation of 30% (memories backing each episode's intent divided by six, episode-weighted). The conditional split is stark: when the model's first-pass intent was promotional pricing — the action the corrupted memory backs — it verified that memory in **48/48** of episodes. When its intent was anything else, **136/252**. Pricing-intent episodes arise only in three of the six models (haiku-4-5 (10), sol (8), terra (30) episodes), so the pooled contrast partially reflects model composition; stratifying by model, the effect survives cleanly within the models that ever intend pricing — Haiku 10/10 vs. 11/40, Terra 30/30 vs. 15/20, Sol 8/8 vs. 38/42 — Cochran–Mantel–Haenszel $p = 2.4 \times 10^{-6}$ (the naïve pooled Fisher test gives $p = 4.5 \times 10^{-12}$ and overstates the evidence). Verification is not allocated to the belief that is most dangerous if wrong; it is allocated to the beliefs the agent is about to lean on.

4.2 The corruption is detectable, unevenly (H2)

Pooled over models, the pricing memory attracts verification in 184/300 of drifted episodes versus 38/150 of clean ones (CMH stratified by model $p = 6.9 \times 10^{-15}$; pooled Fisher $p = 2.6 \times 10^{-12}$): deleting the caveat makes the memory *more* likely to be checked, so the reallocation is a response to the belief being under-specified rather than noise. But Table 2 shows two facts a pooled number hides. First, per-model drifted rates at the identical task span Luna's 20% [11, 33] to Sol's 92% [81, 97] (Wilson 95% intervals) — Luna leaves the corrupted belief unexamined four times in five while Sol checks it almost always, and this heterogeneity does not track any obvious capability ordering. Second, one model *reverses* the direction: Sonnet 5 verifies the pricing memory more often when its caveat is *present* (76%) than when it is absent (52%; the reversal is marginal at $p \approx 0.05$). Sonnet's first-pass intent is onboarding in every one of these episodes in both arms, so the shift is purely one of verification allocation, not of plan: for Sonnet, a visible warning acts as a verify-me flag, and removing the warning removes the flag — a reminder that "caveat present" is not a neutral control condition for every model, and a second mechanism by which silently dropped caveats evade review.

4.3 The same corruption is caught when it lies in the agent's path (H3)

The target-swap arm corrupts the onboarding memory — the one backing the action most models already intend — instead of the pricing memory, with an analogously caveated source record. On the three ablation models, the corrupted memory is verified in **75/75** of intent-aligned episodes, against **92/150** when the corrupted belief is intent-misaligned ($p = 1.4 \times 10^{-12}$). Every model, including Luna at 25/25 aligned versus 10/50 misaligned, catches the aligned corruption every single time. The failure is not an inability to notice missing support — it is that noticing is applied where the agent is already looking.

4.4 Instructions raise the floor, not the ceiling (H4)

The triage invariant — one sentence naming no memory, metric, or direction — raises verification of the corrupted memory in every model: Opus 36/50 → 49/50, Sonnet 26/50 → 49/50, Haiku 21/50 → 44/50, Sol 46/50 → 50/50, Terra 45/50 → 48/50, Luna 10/50 → 37/50. Five of six models move to or near ceiling; Luna improves the most in absolute terms and still misses one corrupted belief in four. Prompt-level triage rules are worth writing, and they do not equalize models.

4.5 Budget buys direction (H5)

At budget one, the ablation models verify the corrupted belief in 4/25 (Opus), 16/25 (Sol) and 0/25 (Luna) of drifted episodes — the single credit goes to the intended action's own evidence almost by default. At budget three the rates rise to 24/25, 25/25 and 17/25. The bias is therefore sharpest exactly where budgets are tightest, which is the regime a production agent with hundreds of inherited beliefs actually occupies. Figure 1 shows the three curves.

4.6 Controls

Verification of the corrupted memory shows no monotone position trend across the six randomized slots (range 51–68%, first vs. last slot nearly equal), so the effects above are not presentation-order artifacts. Reversals follow verification: in drifted episodes where the corrupted source was retrieved, models changed their intended action or its scale 74/184 of the time (40%) after reading it, which is what makes the allocation decision consequential: the retrieved caveat is acted on when it is retrieved at all.

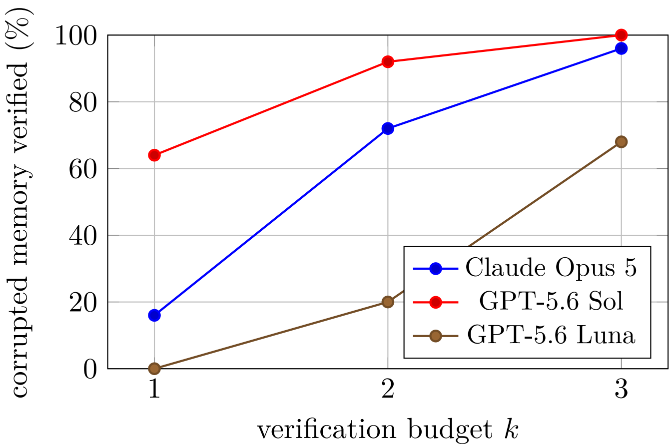

Figure 1: Probability that the corrupted memory's source is requested, by verification budget (drifted arm, intent-misaligned target, n=25 or 50 per point). The allocation bias is sharpest at the tightest budget.

4.7 Why intent-aligned allocation is locally rational — and when it stops being so

Within a single decision, Experiment 1's allocation is defensible. Let A be the intended action and m_A the memories backing it. Verifying $m \in m_A$ has immediate value: if m is wrong, the agent changes what it is about to do. Verifying a belief behind an action the agent is *not* taking has, within the episode, an expected value of zero — the belief is never exercised. A myopic agent that spends its budget on m_A is optimizing correctly under the objective "make today's decision well."

The value of verifying memory m is better written

$$\mathrm{EVOI}(m) \;=\; \textstyle\sum_c P(c)\, P(m \text{ load-bearing in } c)\, P(m \text{ corrupted})\, L(m, c),$$

where c ranges over the *future contexts the belief will survive into*, and L is the loss averted by catching the corruption before acting in c. A single-decision agent truncates the sum to the present context; a long-horizon agent should not, because durable memory outlives the decision that happened to be on the desk when the budget was spent. Experiment 2 makes one of the truncated contexts arrive.

4.8 Experiment 2: the unexamined belief becomes load-bearing

Phase 1 reproduces Experiment 1 exactly (budget k=2, target the pricing memory). Then, in the same session, the situation shifts: the competitor cuts prices again, churn among price-sensitive SMB accounts doubles, and leadership wants a response *today* — with the verification budget spent. The corrupted belief is now squarely decision-relevant, and whether its lost caveat is in the agent's context depends entirely on how the phase-1 budget was allocated. We ran 450 two-phase episodes (six models × three arms × 25) plus 150 single-call audit probes, pre-registered as H6–H8 with scoring fixed before any call. Phase-2 verification requests are declined by design; 60 of 450 phase-2 answers still listed lookup ids, which were not honored.

The pre-registered failure metrics returned zero — and the reason is itself a finding. Pre-registered *reliance* (pricing chosen, corrupted memory cited, no uncertainty preserved) fired in 0/150 drifted episodes; the strict *harmful* rule in 0/150. As specified, H6 and H7 are **not**

supported. The decomposition explains why: of the 50 drifted episodes whose phase-2 choice *was* promotional pricing, all 50 cited the corrupted memory as evidence — and all 50 also set the uncertainty flag, 49 flagged downstream risk. Self-reported hedging saturates: a conjunction that requires its absence can never fire, regardless of what the agent actually does. Behavioral fields tell a different story, and everything that follows is **exploratory**, stated as such.

Exploratory: what the agent knows at phase 2 governs where it walks. In the drifted arm, 50/150 of phase-2 decisions chose promotional pricing. Conditioned on the phase-1 budget: episodes whose budget reached the corrupted memory — putting the recovered source record, caveat included, into context — chose pricing 6/94 of the time; episodes whose budget missed it, 44/56 (CMH stratified by model, $p = 1.3 \times 10^{-14}$). The direction is the same in *every* model (e.g., Luna 18/18 unverified vs. 1/7 verified; Haiku 16/16 vs. 3/9; Sonnet 5/9 vs. 0/16). Because verification is self-selected, we also use the randomized contrast: the drifted-triage arm differs from drifted only by the one-sentence invariant, and its phase-2 pricing rate is 21/150 against 50/150 (CMH $p = 8.7 \times 10^{-6}$). A rule written to steer *verification* halved a *behavioral* rate one decision later.

The only non-guarded pricing actions in the study occur here: 9 in the drifted arm (of which 3 remain under triage) against 0/150 in clean (Fisher $p = 0.003$). In the clean arm, 41/150 phase-2 decisions still chose pricing — with the caveat visible, every one of them as a guarded test with downstream risk flagged. Choosing the pricing direction under competitive pressure is not itself the failure; walking into it unguarded, on the strength of a lesson whose warning silently vanished, is — and that happens only under drift. We also note, without a clean within-arm comparison to back it, that a *retrieved* source record suppresses the corrupted direction far more strongly (6/94) than the same caveat sitting passively in inherited memory (41/150 choose pricing); recency and salience of provenance, not mere availability, may be doing the work.

H8: stated support judgments dissociate from allocation — in both directions. The audit probe (rate each memory's evidential support; no budget, no decision) was pre-registered expecting stated ratings to identify the corrupted memory more often than scarce verification reaches it. That holds for exactly one model — Haiku names it least-supported 21/25 while verifying it 21/50 — and is *reversed* for the strongest verifiers: Sol names it 0/25 yet verifies it 46/50; Opus 1/25 vs. 36/50; Terra 1/25 vs. 45/50 (these three overwhelmingly rate the vaguest memory, activation messaging, as least supported). H8 as pre-registered is **mostly not supported**; what the probe reveals instead is a double dissociation: models that verify the corrupted memory do not describe it as weak when asked, and the model that describes it as weak does not spend its budget on it. Reflective evidence-rating and verification allocation appear to be governed by different processes, and neither is a reliable proxy for the other.

5 Limitations

One scenario, one domain. Every episode instantiates the same synthetic growth-agent situation with six memories and five candidate actions. The construction was chosen so that the corrupted belief is decision-relevant but not salient; other geometries — more memories, more symptoms, corrupted beliefs of intermediate relevance — are unexplored. Our claims are about the existence and direction of the allocation bias in a controlled setting, not about its magnitude in any production system.

Single-decision episodes. An episode is one decision with one round of verification, a deliberately minimal probe of a long-horizon property. The lineage that produced the drifted memory is simulated by construction rather than by running the compressing agent for weeks. Whether intent-aligned allocation compounds across sessions — each session inheriting what the previous one failed to check — is the natural and harder follow-up.

The pilot's headline estimate moved under randomization. Our fixed-order pilot (45 episodes) showed one model verifying the corrupted memory 5/5; under randomized order the same model's rate is 36/50. With n=5 the pilot cannot statistically distinguish these (Fisher $p = 0.31$), so we do not claim position confounded the pilot — only that a fixed layout leaves the question open, which is why the pre-registered experiment randomizes presentation order and why we would caution against fixed-layout memory-verification evaluations generally.

Harmful-decision rule is strict by design. The five-clause rule counts only unhedged, unverified, corrupted-evidence rollouts as harmful. Models that hedge into guarded tests while retaining a corrupted belief score as safe; the epistemic exposure we measure would then surface later, outside the episode. Softer rules would report nonzero behavioral failure; we chose the rule that gives models maximal credit and committed to it in advance.

Verification is answered truthfully and instantly. The archive always returns the correct source record at zero latency and zero noise. Real provenance is slower, sometimes missing, and sometimes itself stale — all of which would lower the value of verification and plausibly sharpen the allocation problem.

Model coverage and settings. Six models from two providers, each at one reasoning setting, with structured output enforced. Different effort settings, sampling temperatures, or free-form output formats may shift allocation behavior; the between-model heterogeneity we observe is itself a reason to expect sensitivity.

Self-reported hedges saturate. Experiment 2's pre-registered reliance and harm rules required the absence of a self-reported uncertainty flag, and every drifted episode that chose the corrupted direction set that flag; the pre-registered metrics therefore returned zero by construction, and we say so rather than substituting a post-hoc rule silently. All Experiment-2 findings beyond that null are labeled exploratory. Behavioral fields (the chosen scale; the chosen action) carried the signal instead; future designs should pre-register behavioral hedging measures and treat introspective flags as unscoreable.

The audit probe measures stated judgment, not necessarily belief. H8's probe asks models to rate evidential support from the text alone. Strong verifiers rated the assertive-but-corrupted memory well-supported while spending budget on it anyway; the vaguest memory drew their lowest ratings. The probe may therefore measure surface assertiveness of the written lesson rather than the omission-sensitivity that drives allocation, and the double dissociation we report should be read with that construct caveat.

Episode files store seeds and order, not raw prompt text. Episode records contain the seed-derived presentation order and all structured answers; the exact prompt is reconstructible deterministically, and the seeded shuffle reproduces the stored order in 1,620 of 1,620 files across both experiments. Raw prompt text embedded per episode would have been better practice, and Experiment 2's phase-two lookup denials (60 of 450 answers still requested lookups) are disclosed rather than hidden.

6 Conclusion

Long-horizon reliability work has concentrated on whether agents can detect corrupted state, and on how to spend verification compute across candidate outputs. Between the two sits a quieter decision that our measurements suggest does much of the work: with limited capacity, an agent verifies where it already intends to act. On a task where nearly every episode spent its full budget (1017 of 1020 in Experiment 1), reliability differences came almost entirely from *direction* — whether scarce attention reached the belief that was silently wrong, or only the beliefs behind the plan the agent had already formed. That allocation is locally rational and, our second experiment suggests, exactly backwards once the context shifts: when the corrupted lesson later came under pressure with the budget spent, what the earlier lookups happened to recover was the difference between walking into the corrupted direction 6% of the time and 79% of the time — in every model we tested — and a one-sentence rule that steers *verification* halved that *behavioral* rate one decision later. Meanwhile stated evidence judgments tracked neither: the strongest verifiers did not call the corrupted memory weak, and the model that called it weakest barely checked it. Verification allocation is its own faculty — distinct from detection, from stated judgment, and from capability — and making it visible, steerable, and eventually risk-weighted rather than intent-weighted is a concrete, testable target for agent-memory systems.

References

Pritam Dash, Tongyu Ge, Aditi Jain, et al. From untrusted input to trusted memory: A systematic study of memory poisoning attacks in llm agents. *arXiv preprint arXiv:2606.04329*, 2026.

Zhengru Fang, Senkang Forest Hu, Zhonghao Chang, et al. Inference-time budget control for llm search agents. *arXiv preprint arXiv:2605.05701*, 2026.

Joshua Klayman and Young-Won Ha. Confirmation, disconfirmation, and information in hypothesis testing. *Psychological Review*, 94(2):211–228, 1987.

Junchi Liao. Auditing provenance sensitivity in llm agent action selection. *arXiv preprint arXiv:2607.20827*, 2026.

Yuxiang Lin, Zihan Wang, Mengyang Liu, et al. Bagen: Are llm agents budget-aware? *arXiv preprint arXiv:2606.00198*, 2026.

Pranav Singh. When does belief-based agent memory help? reliability-conditional updating and provenance-capped poisoning defense. *arXiv preprint arXiv:2606.22030*, 2026.

Gou Tan, Zhensu Sun, Jieke Shi, et al. Agentchaos: Chaos engineering for agent systems via programmatic fault injection. *arXiv preprint arXiv:2608.06790*, 2026.

Daniel Wang and Andrew Xu. Allocbench: Measuring online tool allocation capability in llm agents. *arXiv preprint arXiv:2607.23332*, 2026.

Yiqi Wang, Jiaqi Zhang, Taotao Cai, et al. From agent traces to trust: A survey of evidence tracing and execution provenance in llm agents. *arXiv preprint arXiv:2606.04990*, 2026.

Peter C. Wason. On the failure to eliminate hypotheses in a conceptual task. *Quarterly Journal of Experimental Psychology*, 12(3):129–140, 1960.

Weiwei Xie, Shaoxiong Guo, Fan Zhang, et al. Memevobench: Benchmarking safety risks from memory misevolution in llm agents. *arXiv preprint arXiv:2604.15774*, 2026.

Jinlong Yang. Heteroskedastic signals in budgeted llm verification: Structural heterogeneity limits optimization gains. *arXiv preprint arXiv:2606.15841*, 2026.

Wenhao Yuan, Chenchen Lin, Jian Chen, et al. Belief-guided inference control for large language model services via verifiable observations. *arXiv preprint arXiv:2604.27536*, 2026a.

Wenhao Yuan, Chenchen Lin, Jian Chen, et al. Verify before you commit: Towards faithful reasoning in llm agents via self-auditing. *arXiv preprint arXiv:2604.08401*, 2026b.

A Scenario text

The complete system prompt, memory lineages for both corruption targets, the hidden source records, and the pre-registered scoring rules are released verbatim in the repository (`runs/paper/preregistration` and are byte-identical to what every model received.